



اَوْنُوْرَسِيَّتِي تِيَكْنُوْلُوْجِي مَارَا
UNIVERSITI
TEKNOLOGI
MARA

ICT602

Mobile Technology and Development

TOPIC 1 – History of Mobile Computing

LESSON OUTCOME



Identify, explain and Discuss:

1. Describe Wireless Technology and its history
2. Describe Mobile Computing Concept and User Characteristic
3. Explain Mobile proliferation



Wireless Technology

Telephony and Messaging Concept

Mobile Computing Concept



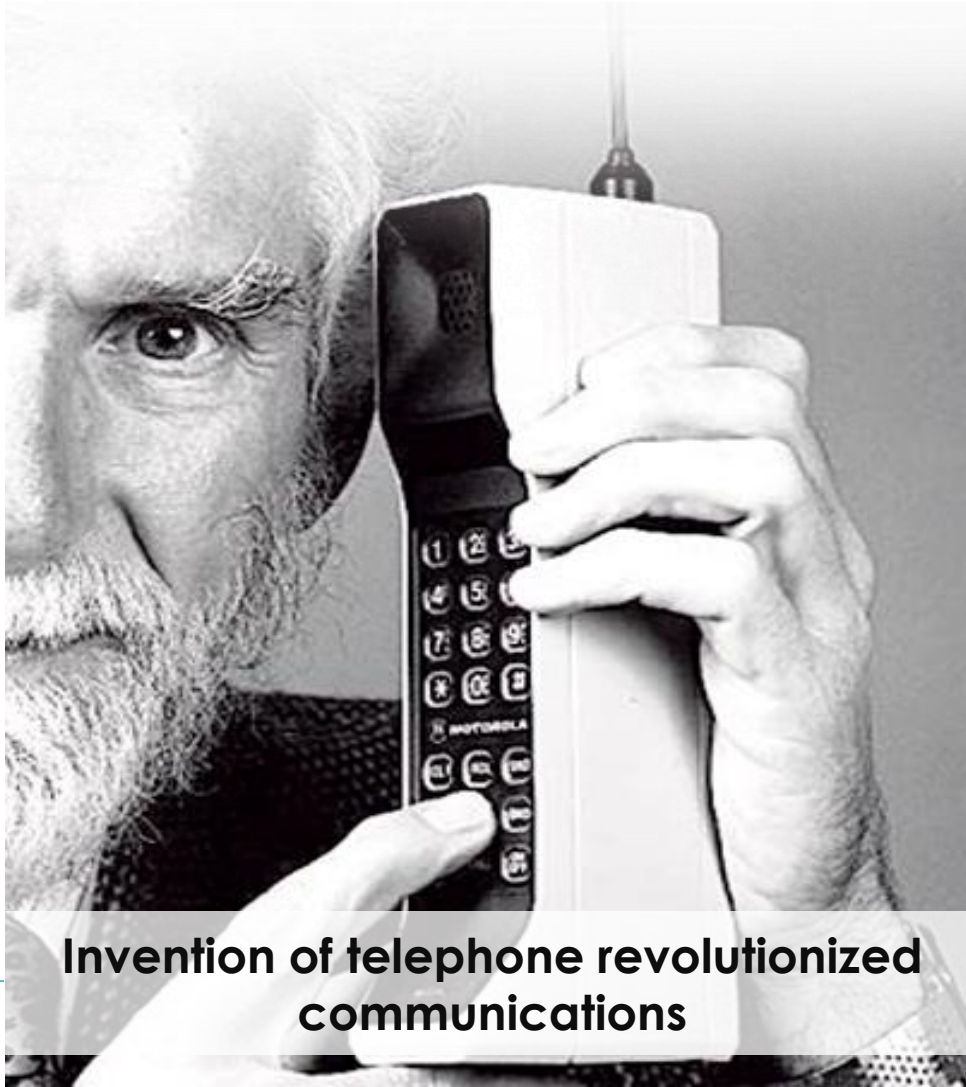
WIRELESS TECHNOLOGY



In the Beginning



The telephone is one of the greatest inventions of mankind



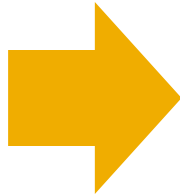
Invention of telephone revolutionized communications



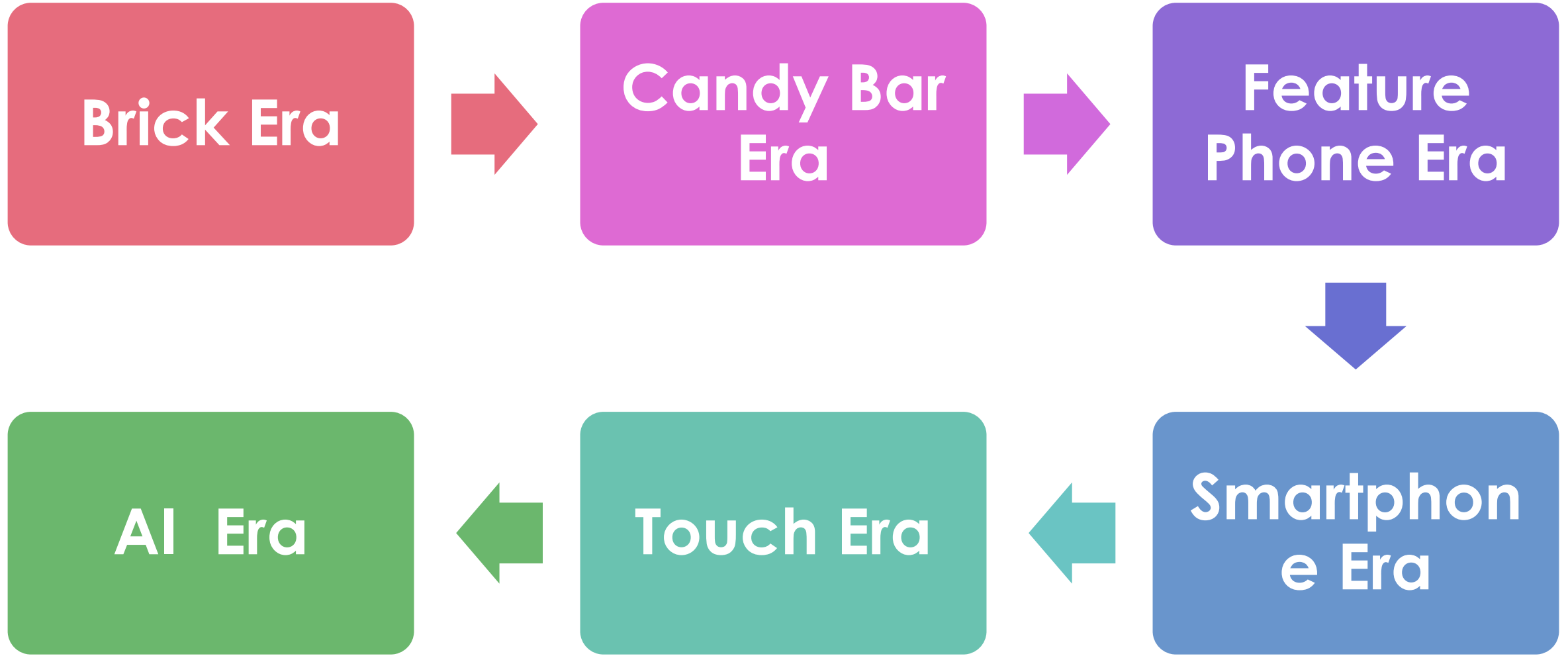
Wireless communication furthers phone communication over electromagnetic signal leading to cellular and mobile device



Evolution of Devices



Evolution of Mobile Devices



Brick Era

- ▶ 1st generation of mobile device (1973-1988)
- ▶ Bulky, has chord.
- ▶ Has **large batteries**
 - ▶ **require enormous power** to reach the nearest **cell tower site**, which were **few and far between**
- ▶ Only provide **analog voice communication**
- ▶ **expensive** and only used by specific group of people



Motorola
DynaTAC 8000X



MOTOROLA
DYNA T-A-C 8000X
CELLULAR PORTABLE TELEPHONE

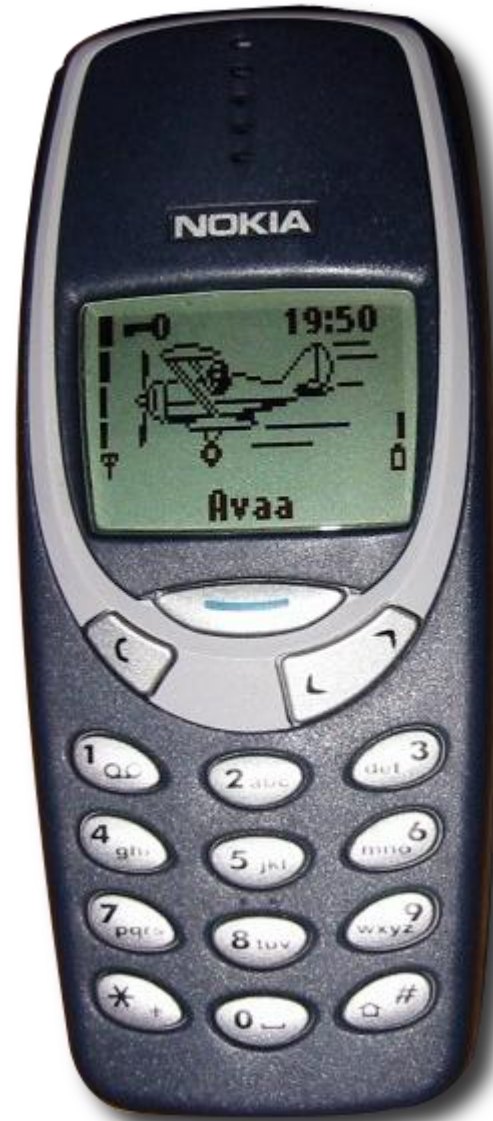


Candy Bar Era

- ▶ Candy Bar is a term to describe the long, thin, rectangular form factor of the majority of mobile devices used during the era (1988-1998)
- ▶ Support **2G technology** and SMS.
- ▶ The phone are **significantly smaller**.
 - ▶ Number of cell towers exploded by increased demand, has reduced the device power requirements.
- ▶ more affordable to the masses



A Nokia candy bar phone



Falls to the floor



Break the screen

Falls to the floor



Break the floor





- × No GPS
- × No Social Apps
- × No Chatting Apps
- × No virtual keyboard



Feature Phone Era

- ▶ 1998-2008
- ▶ Incorporation of **computing power**
- ▶ Allows variety of use and services on the phone such as **listening to music, taking photos and introduces internet** on a phone.
- ▶ Feature Phone Era introduces the concept of **customizable content**

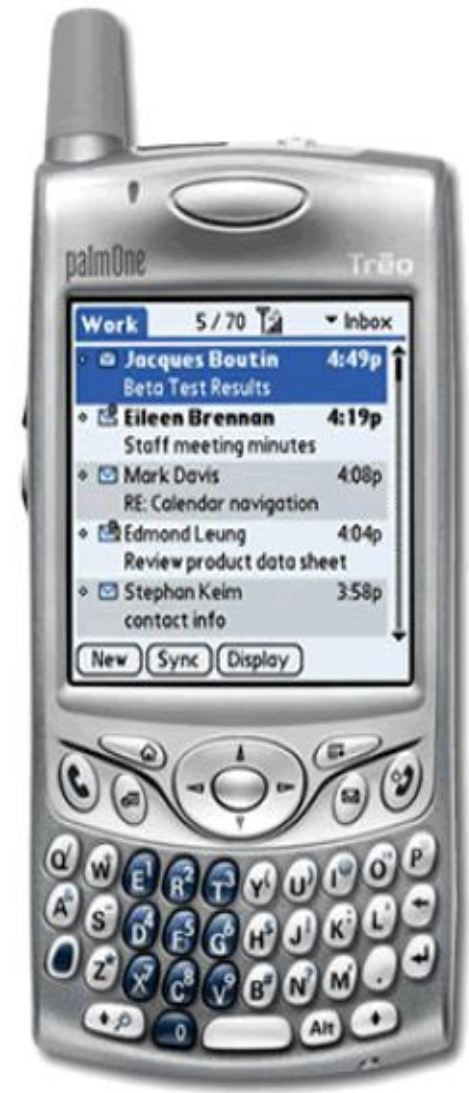


Motorola RAZR



Smart Phone Era

- ▶ Occurred at the same time as the Feature Phone and Touch Era (2007-now?)
- ▶ Incorporate **multi-tasking** OS
- ▶ Shares common OS - Applications developed for smartphone often can be shared across devices
- ▶ **3G** technology



Early smartphones

Touch Phone Era

- ▶ The Touch Era beginnings is marked by the launch of the first iPhone back in 2008.
- ▶ Support **multi-touch touchscreen**.
- ▶ Touch Era phone also includes a standard **set of sensors**



AI Integration Era (2020-present)

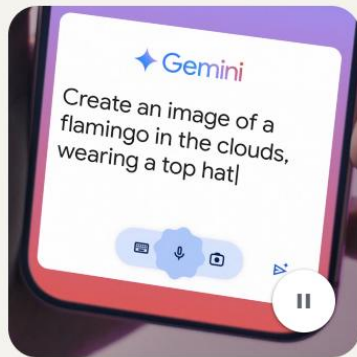
- ▶ This era marked by the integration of AI agent in mobile devices
- ▶ The inclusion of a Neural Processing Unit (NPU), which is capable of parallel tensor processing, significantly enhances on-device AI performance
- ▶ NPU example: Samsung Exynos, Apple Neural Engine (A15 Bionic), Google Tensor G3 processors.



AI Integration Era (2020-present)



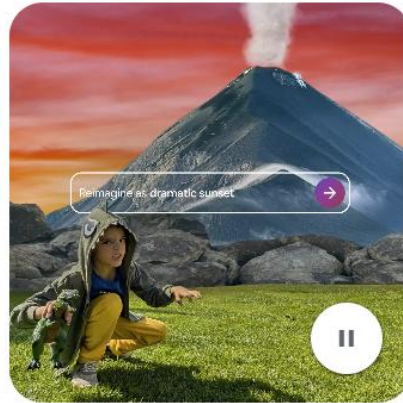
- AI-Driven Functionalities
- Integration of Small Language Models (SLMs)
- Personalization and Context Awareness
- Advanced Camera Systems with AI Processing



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Circle to Search. Whoa.

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Evolution of Devices



**MOBILE PHONE CAN RIVAL DESKTOP PC
IN TERMS OF FUNCTIONALITY**



**GREATER
PROCESSING POWER**



**TOUCHSCREEN
INTERFACE**



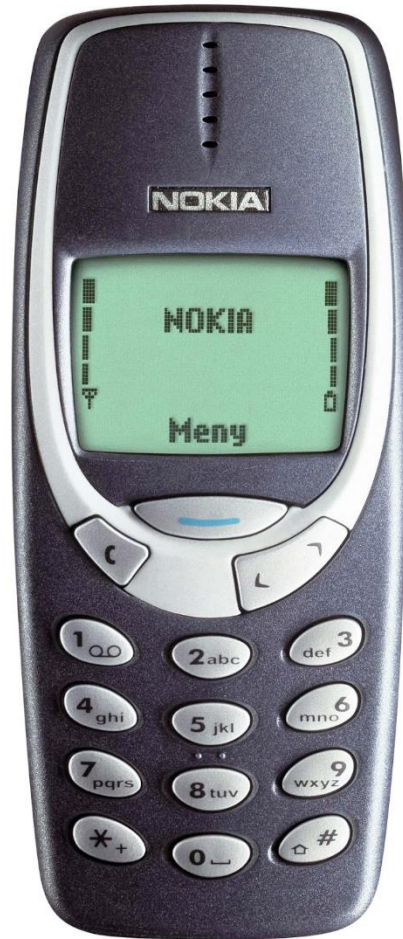
**HIGHER
RESOLUTION**



Your first mobile phone?



- ▶ Do You Remember Your First Cell Phone? When?





TELEPHONY AND MESSAGING CONCEPT



Network Generation



- ▶ 1G ~ 5G
- ▶ Each Generation is defined as a set of telephone network standards , which detail the technological implementation
- ▶ The aim of wireless communication is to provide high quality, reliable communication just like wired communication
- ▶ Each new generation of services represents a big step in that direction

1G, 2G, 3G Networks



Circuit-switched,
analog signals,
Analog Voice only

1G

Circuit-switched,
digital signals, voice
or data overlay, 9
kbps or 19 kbps

2G

Supports **faster
wireless data
services**, GSM
extensions.

**2.5G
GPRS**

Wider bandwidth
and higher data
rates for mobile
users.

**3G
&
4G**

Packet-switched,
Transparent
roaming, 2Mbps,
Identification of
caller location



1G

1ST GENERATION *wireless network*

- Basic voice service
- Analog-based protocols



2G

2ND GENERATION *wireless network*

- Designed for voice
- Improved coverage and capacity
- First digital standards (GSM, CDMA)



3G

3RD GENERATION *wireless network*

- Designed for voice with some data consideration (multimedia, text, internet)
- First mobile broadband



4G

4TH GENERATION *wireless network*

- Designed primarily for data
- IP-based protocols (LTE)
- True mobile broadband



T H E N E E D F O R S P E E D *in kilobits per second*

2.4 *kbps*

64 *kbps*

2,000 *kbps*

100,000 *kbps*



1G Network



- ▶ **Analog technology**
- ▶ poor battery life and voice quality
- ▶ less secure
- ▶ experience **dropped calls** .
- ▶ The first commercial cellular network was introduced in the late 70's
- ▶ The United States has approved the first 1G operations with Motorola DynaTAC and Nokia Mobira Talkman becoming the first mobiles phones to see widespread use in that country.

2G Network



- ▶ **Digital signal.**
- ▶ provide secure and reliable communication channel.
- ▶ It implemented the concept of **CDMA** and **GSM**
- ▶ The **use of data also along with voice.**
- ▶ New features - SMS, internal roaming, conference calls, call hold and billing based on services
- ▶ Initially data transfer speed is 9.6kbit/sec. The maximum speed with **GPRS** is 50 Kbps or 1 Mbps with **EDGE**.

3G - Third Generation



- ▶ Utilises a new technology called **Universal Mobile Telecommunications System** as its core network architecture.
- ▶ Based on a set of standards used for mobile devices networks that comply with the International Mobile Telecommunications-2000 specifications by the **International Telecommunication Union**
- ▶ 3G has Multimedia services support along with streaming are more popular.
- ▶ 3G increased the efficiency of frequency spectrum by improving how audio is compressed during a call, so **more simultaneous calls** can happen in the same frequency range.
- ▶ The UN's International Telecommunications Union **IMT-2000** standard requires **stationary speeds of 2Mbps and mobile speeds of 384kbps for a "true" 3G.**

4G – Fourth Generation Network

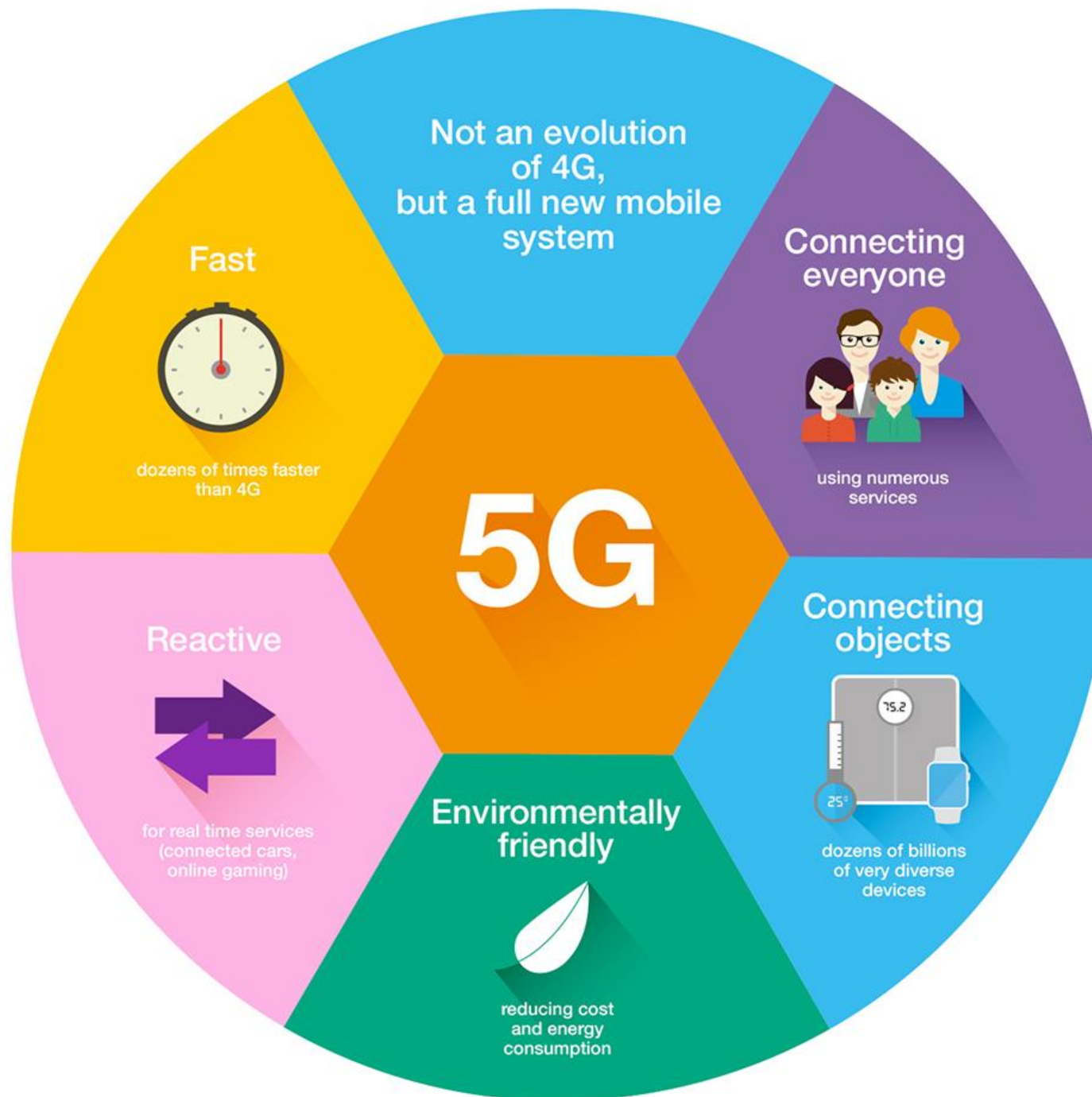


- ▶ 4G is based on packet switching only.
- ▶ Also known as Long-Term Evolution (LTE) standard.
- ▶ first commercially deployed in Oslo, Norway and Stockholm, Sweden in 2009
- ▶ Provide high speed , high quality and high capacity while improving security and lower the cost of voice and data services and multimedia applications
- ▶ Maximum speed of a 4G network when the device is **moving is 100 Mbps / when stationary is 1 Gbps**
- ▶ 4G devices also offers Voice over Long-Term Evolution or **VoLTE**

5G – Fifth Generation Network



- ▶ **data speed from 50Mbps to more than 4Gbps**
- ▶ 5G reduces the latency 30 milliseconds in early deployment stage.
- ▶ The standards bodies involved are **aiming for 5G to delivery data at 20Gbps speed and 1ms latency**,
- ▶ 5G networks use a type of encoding called OFDM, which is similar to the encoding that 4G LTE uses.
- ▶ 5G generations are designed to cater for low power devices and aims to support AR/ VR applications.





POTENTIAL USES OF 5G



Cloud-based systems will be able to stream software updates, music, navigation data and traffic conditions to driverless cars



Download time for a high-definition full-length movie will be seconds, not minutes



5G speeds offer the potential for simultaneous language translation between people attending a teleconference

SMS: Short Message Service



allows short text messages to be sent between mobile devices.

non-voice communications.

SMS loses value as latency increases



Typical SMS Applications



CONSUMERS

1. Person to person messaging
2. Interactive Information services
3. Entertainment services

CORPORATE

1. Notification & alert services
2. Managing contacts & Appointments
3. Content Push

SMS Advantages: Is it still relevant?



- ▶ Support modern and older devices, including embedded device.
- ▶ Almost guaranteed to work on any mobile network.
- ▶ Can be used to send text messages programmatically
- ▶ Reliable and can guaranteed to work on any mobile devices.
- ▶ SMS messages can be sent to devices without internet connection
- ▶ SMS messages can be sent even if the receiver's device is switched off.
- ▶ Supports sending to multiple receivers and supports broadcast.
- ▶ Can be used in Application-to-Person (A2P) or Machine-to-Machine (M2M) applications



Wednesday, 7 Dec 2022 • 15:17

RMO MySejahtera: You are COVID-19 positive. Kindly refresh your MySejahtera Profile and perform home monitoring assessment in MySejahtera.

Risks and Attack Associated with SMS



- ▶ SMS Spoofing
- ▶ SMS Phishing
- ▶ SMS Spam
- ▶ SMS Subscription Fraud
- ▶ Denial of Service Attack

MMS: Multimedia Messaging Service



- ▶ The Multimedia Messaging Service (MMS) has the ability to send and receive messages comprising a combination of text, sound, images and video over mobile devices
- ▶ MMS is a more advance version of the current SMS (Short Messaging Service)
- ▶ MMS is a standard set by 3rd Generation Partnership Project (3GPP)



Unpopularity of MMS?

- ▶ High Cost
- ▶ Slow Delivery
- ▶ Compatibility Issues
- ▶ Emergence of Instant Messaging Apps



RCS: Modern Messaging Standard



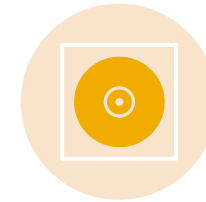
- ▶ RCS – Rich Communication Services is defined by GSMA
- ▶ Designed to replace SMS and MMS using IP-based messaging
- ▶ RCS is designed to blend with SMS sending applications in modern mobile devices.



RCS improved upon SMS/MMS :



Removed SMS characters limitation (160 chars)



Improved multimedia capabilities, compared to MMS



Cheaper to send messages / IP-based.



Offer graceful fallback to SMS to unsupported device.



RICH COMMUNICATION SERVICES THE NEXT-GENERATION OF SMS

FOREST
INTERACTIVE



RCS is an **IP-based messaging** service, which is provided to users through mobile operators.



RCS incorporates the global reach of SMS with the rich features of chat apps to deliver **enhanced user experiences**.



RCS Business Messaging enables brands to **engage with customers** by using chatbots or AI.



88 Operators launch RCS in 59 countries by 2020.



Estimated **390 million** monthly active users worldwide.



Estimated value of business of **\$74 Billion** by 2021.



1 Verified Sender

2 Customized Branding
Custom name, color & logo/icon

3 Rich Media
Images, videos, GIFs

4 Suggested Actions

© GSMA Intelligence

RCS: Modern Messaging Standard

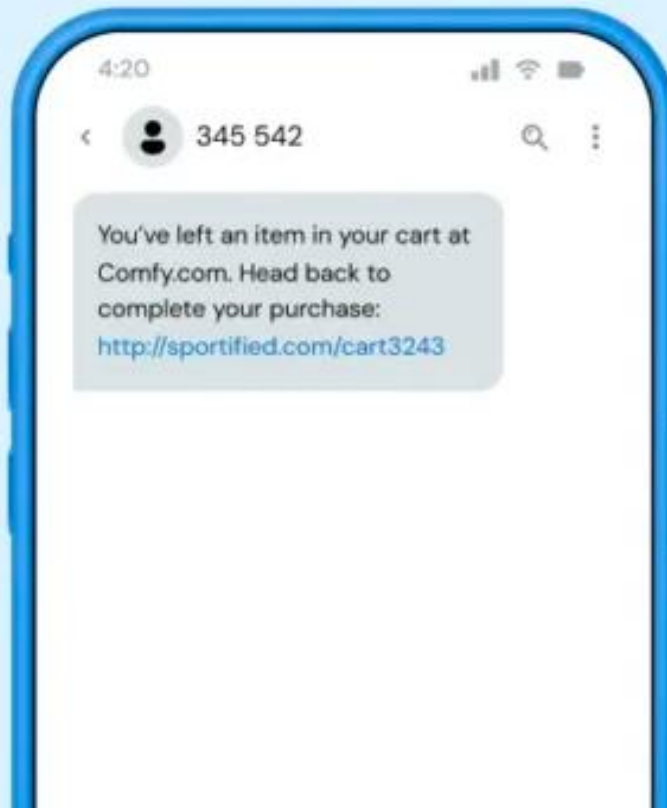


Key RCS features:

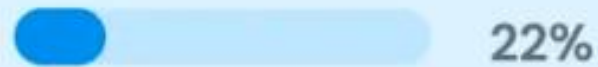
1. Typing indicator
2. Reliable read receipts
3. End-to-end encryption
4. Group messaging
5. Improved multimedia capabilities
6. Integration into multiple application
7. Supported standard by GSMA

Which messaging experience do consumers prefer?

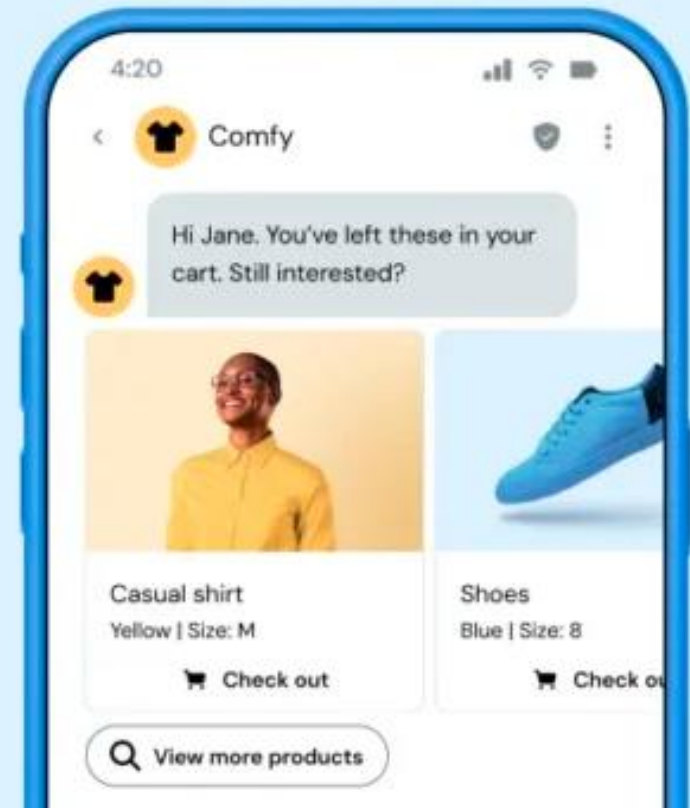
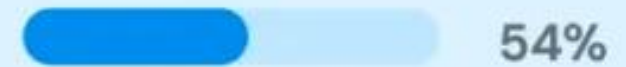
SMS



MMS



RCS





MOBILE COMPUTING CONCEPT





DIVIDED INTO DEVICE CHARACTERISTICS AND USER CHARACTERISTICS:

Mobile Device Characteristic

- **Portable**
- **Connected**
- **Limited Storage**
- **Limited Power**
- **Limited Resource**

Mobile User Characteristic

- **Mobile**
- **Interruptible, Easily Distracted**
- **Available**
- **Sociable**
- **Identifiable**

Portable



Mobile device
are designed
to be **movable**
and portable

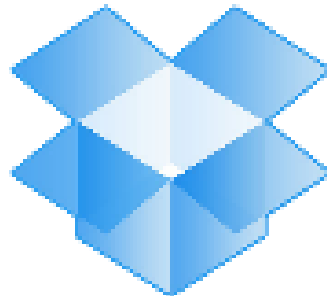
It is meant to be
carried by its user
and **kept close**
by



Connected



- Mobile devices are meant to be connected and are expected to be **connected to any form of network**
 - Even the most primitive form of mobile device is connected to a GSM network
 - A mobile device without connection is useless
-
- Users expect to use mobile device to connect with other users.



Limited Storage



Due to its nature, mobile device will always have relatively **limited storage**.



Early phones only have limited space to store contact names and phone numbers

Later candy-bar era phones allows users to store notes and reminders in a built in memory



Limited Storage



Even by current standards, mobile phones still in need of 'extra' storage.

Storage space is still one the deciding factor in buying phones.

Application designed for mobile device should always minimize the use of storage or at least offload the storage to a remote location

Limited Power



One can argue that there is no 'unlimited' power

Power is significantly limited by its battery capacity

Factor that affects battery drains:

Weak cell tower signal

Extensive processing cycles

Extensive use of sensors

Connectivity

Screen activity

Background process activity

A good application should keep battery drains to a minimum level

Limited Resource



Resource in mobile device refers to its memory, processing capabilities and connectivity.

Memory (or RAM) is limited in mobile device. Applications should reuse objects and discard objects that are out of scope to conserve memory. Devices will often stalled when it runs out of memory during use.



Processing power in mobile devices is usually constrained to conserve battery and to reduce heat. Application that uses extensive and prolong computation usually offloads them to remote server.



 **Mobile device has limited connectivity**

 **This is due to:**

Unreliable network coverage

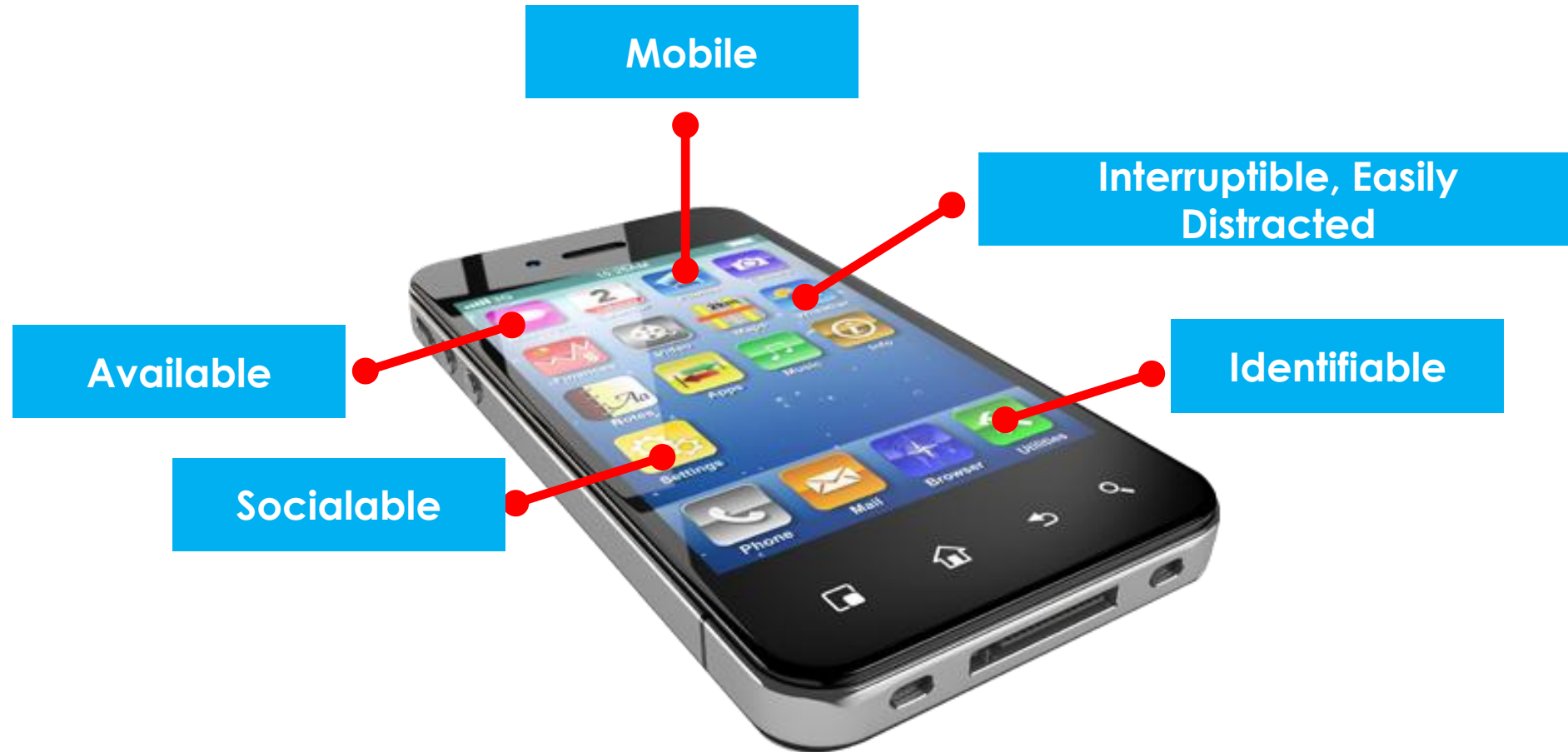
Limited network cell capacity (overcrowding)

Subject to interference (signal bounces off barrier, noise)

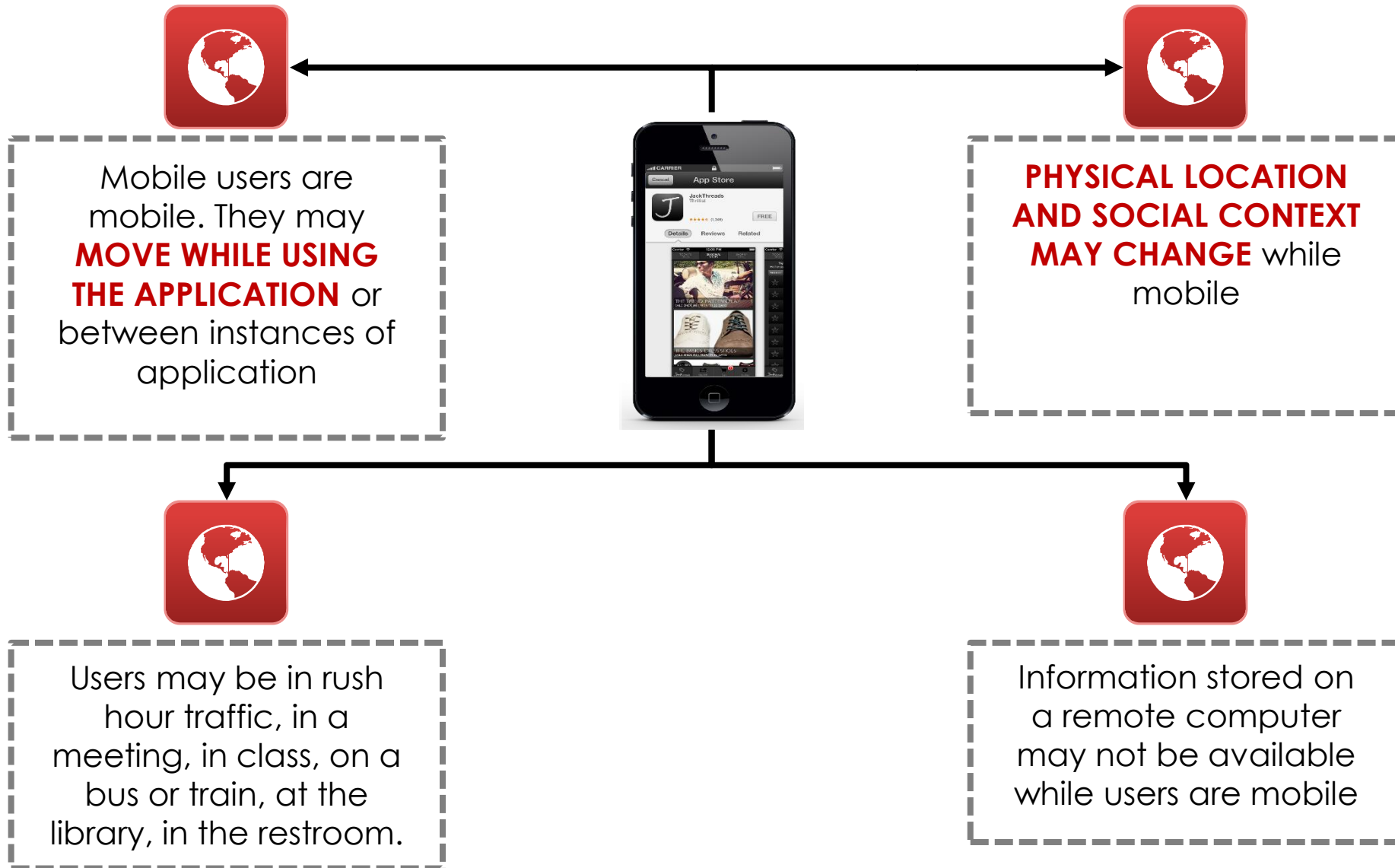
 **Mobile device also suffers from frequent disconnection.**

 **Application designed for mobile device should be tolerant to unreliable speed and connectivity**

Mobile User Characteristics



Mobile



Interruptible, Easily distracted



Mobile users are easily distracted

Users attention may be distracted by his environment or social cues.

Tasks performed on mobile device should always be expected to be interruptible.

Writing SMS or watching videos on mobile can always be interrupted by a phone call.



Available



- ▶ Mobile users are **always available to remote friends and contacts**.
- ▶ The purpose of mobile devices is to make its users 'available'.
- ▶ Mobile devices are likely to be close to users wherever the user goes
- ▶ In fact, Many people feel uncomfortable when separated from their devices



A voice call is socially and technologically assumed between two or more people



Sociability is a key metaphor in mobile app design.



Sociability in mobile phone is a means to reach others around you, to be able to communicate whether verbally or in written form or to immerse in other form of digital social interactions



Users may not be interested to use applications without the element of sociability,

Sociability Concepts Example



1. Social Media Integration Applications
2. Digital Gifting and Reactions in Games and Apps
3. Calendar Sharing and Invitations
4. Photo and Memory Sharing Features

Identifiable



- ▶ Because devices are personal, they are usually **unique to a single user**
- ▶ This identification includes – unique phone numbers, user account, SIM card, email addresses, contact lists.
- ▶ Mobile devices are usually tied to a single user.
- ▶ Therefore, application designed for mobile devices **SHOULD NOT** ask for users for login credential each time when accessing the app.

Identifiable Concept Examples



1. Mobile application do not ask for user login or password everytime you want to use them. You only need to unlock your phone once, then starts using the application.
2. Applications utilizes biometrics sensors (fingerprint, facial recognition or SSO to authenticate users without requiring typing keys.
3. Account registration and creation
4. Personalized User Profiles Based on Device Information
5. Account Recovery via SIM or Phone Number Verification



PLATFORM PROLIFERATION AND DEVICE FRAGMENTATION



Platform Proliferation



- ▶ Mobile devices comes in many **different machine architecture**
- ▶ This lead to **different OS and different machine code binaries** produced for each machine architecture
- ▶ Different manufacturers championed their own OS and set of system libraries for their own devices.
- ▶ Some manufacturers supported several different OS for their devices.

Platform Proliferation



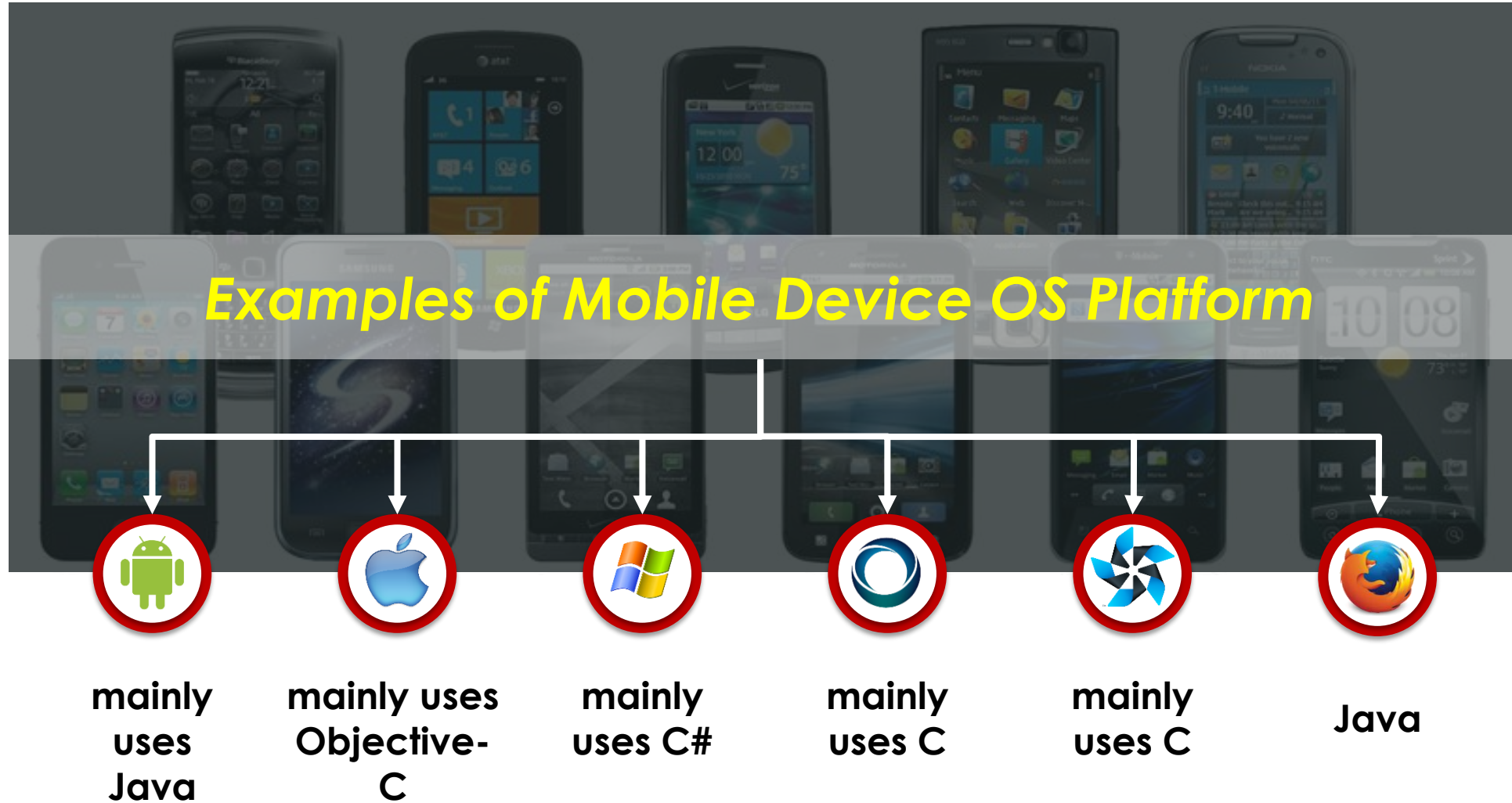
Platform proliferation is unavoidable due to the nature of mobile devices market and the nature of mobile device architecture.

Problem to application developer and service providers.



developers has to support different devices, operating system and device features for the same application.

Platform Proliferation



Platform Proliferation



- ▶ Different platforms available to users means users are divided when buying their device
- ▶ Application **developers may have to choose which platform to support**
- ▶ Some users will not be able to use certain applications which are available on another platform.
- ▶ Complexity of maintaining software increases.

Device Fragmentation



- ▶ A situation where devices for **a platform has vastly different specification, features or capabilities.**
- ▶ Device Fragmentation also refers to device that has **different OS and API**
- ▶ Device Fragmentation can occur within the same platform and is more common in Android and Windows Mobile. Why???



Device Fragmentation



- ▶ Not all mobile devices are created equal
- ▶ This is mainly because of **differences of implementation between platform vendors**.
- ▶ Platform vendors produces wide selection of mobile devices model to **target different part of the market segment**
- ▶ A good developer will design the mobile application which **support the widest range** of devices available

Examples of Device Fragmentation



- ▶ Samsung Galaxy 5 and Samsung Galaxy Note II both are Android smartphones produced by Samsung.
- ▶ Samsung Galaxy Note II has 5.3" screen size, has front and back camera , supports NFC
- ▶ Samsung Galaxy 5 has 2.8" screen size and has a back camera
- ▶ An application developer who releases a "Self-Portrait" camera application must not assume all phones have front an back camera, because doing so may crashes the application on devices like Samsung Galaxy 5.

Examples of Device Fragmentation



- ▶ Ahmad and Siti both use Samsung Galaxy 2 phones, initially the device sold with Android 2.3.7 operating system. However Ahmad decided to upgrade the device OS to Android 4.0.4.
- ▶ One day, Ahmad decides to use a download application which uses WiFi P2P features available in Android 4.0.4
- ▶ Siti saw what Ahmad did and try to load the same application
- ▶ The application does not work on Siti's phone, leading her to speculate that her device might be broken.

Device Fragmentation: Hardware Fragmentation



- ▶ Hardware Fragmentation usually presents on mobile devices as the vendors produces mobile devices with different specification to cater certain market segments.
- ▶ Common market segments:
 - ▶ Entry-Level
 - ▶ Mid-Range
 - ▶ Premium
 - ▶ Flagship
 - ▶ Specialized / Targeted

Device Fragmentation: Hardware Fragmentation



- ▶ An entry-level mobile device usually do not have all the hardware implementation on Premium-class or Flagship mobile device.
- ▶ Common Hardware fragmentation:
 - ▶ Biometric / Fingerprint sensor
 - ▶ Gyroscope
 - ▶ Ambient Light Sensor
 - ▶ Proximity Sensor
 - ▶ Air Pressure Sensor (Barometer)
 - ▶ Grip Sensor
 - ▶ Temperature Humidity sensor
 - ▶ Number of Cameras and Camera Resolution

Examples of Device Fragmentation



- ▶ An application which relies on Fingerprint sensor for authentication must be designed so it could work on devices which does not have Fingerprint sensor to avoid unintentional device lockout.
- ▶ An Entry-Level mobile device without gyroscope would not be able to support application which uses Augmented Reality exclusively.
- ▶ Two devices may have the same 5.5" screen size but may have different resolution pixel-density. This may affect icon sizes and User-Interface element placements.

Device Fragmentation: API Fragmentation



- ▶ API is essentially **a function or method calls made to the OS to perform certain tasks or for accessing hardware sensors** or implementation.
- ▶ API Fragmentation happens when there are changes made to system libraries or API calls which sufficiently **alters its behavior from previous API implementation.**
- ▶ Example: Android frequently documented new API changes which are incompatible with previous API level.

API Fragmentation



- ▶ Abu and Faisal both use Samsung Galaxy A9 which was released in 2019. The device originally shipped with Android 8.0 operating system (API level 26). However, unbeknownst to Abu, his device has been automatically upgraded to Android 9.0 (API level 28)
- ▶ One day Faisal introduced Abu to a third-party location-based game which offers treasure hunts challenges based on user-location.
- ▶ However, Unlike Faisal who can play the location-based game successfully, Abu found out that his mobile device is unable to report nor registers his current location.
- ▶ Thus, Abu thought his device is faulty. Although, in reality there are changes in API level 28 which makes it react differently to certain API calls involving location access.

In short..



- ▶ **Platform Proliferation** means – too many platforms to choose from
- ▶ **Device Fragmentation** - there are many devices with different size, specifications, capabilities and API to target applications for.





Communication technology has evolved over a period of time, influencing how people communicate with each other

Computing and communication technology has combined together to produce mobile and portable devices which has become integral part of our life

This lead to a unique situation where mobile computing users would exhibit distinct characteristic than other type of computing users



LECTURE ENDS

